DataCo Global

Progress Report

Business Intelligence CAPSTONE

3DSA2 - Group 1
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DataCo Global Overview

DataCo Global is the case-study company in the DataCo Smart Supply Chain dataset (Constante, 2019), which covers provisioning, production, sales, and distribution of clothing, sports, and electronic supplies.

Markets
Africa, Europe, LATAM, Pacific Asia, USCA

Payment Types
Cash, Debit, Online Payment, Transfer

Customer Segments
Consumer, Corporate, Home Office

Shipping Modes
Standard, Second, First Class, Same Day

Departments
Apparel, Fitness, Footwear, Golf, Outdoor, Technology

Order Status
Complete, Pending, Canceled, Suspected Fraud

Inside DataCo Global Data

180,519
Rows

53
Columns

2015 – 2018
Order Data Period

Source

DataCo Global is the case-study company in the DataCo Smart Supply Chain dataset (Constante, 2019), which covers provisioning, production, sales, and distribution of clothing, sports, and electronic supplies.

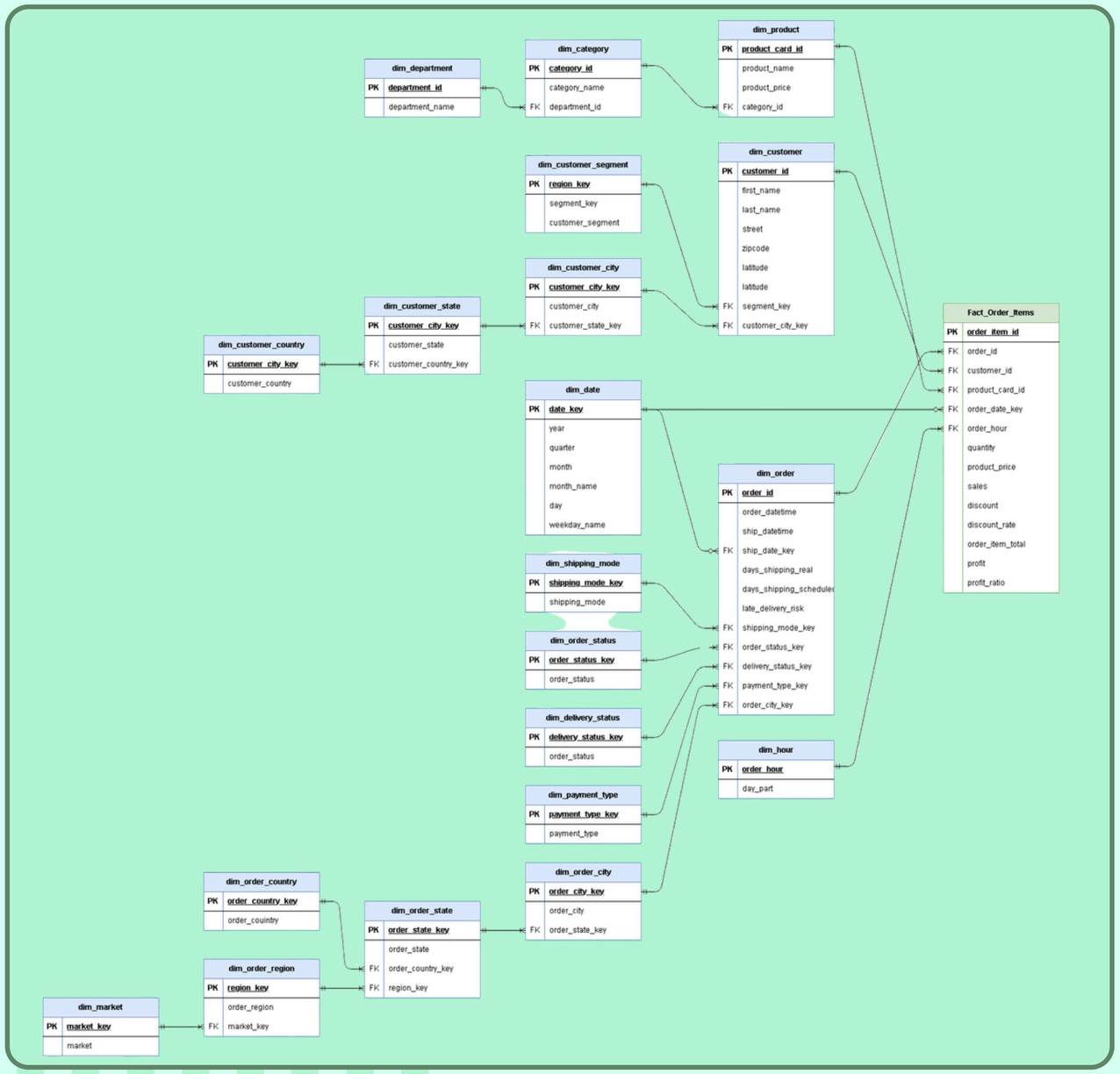
Grain

One Row per Order Item (DataCoSupplyDataChainDataset.csv)

Cleaning

- Loaded using MySQL
- Dates parsed
- Translated Spanish Country Names to English
- Splitted the Data in 21 Tables

Data Source Schema



1	**Fact**	*fact_order_items, one row per order item*
5	**Order**	*dim_order, dim_order_status, dim_shipping_mode, dim_delivery _status, dim_payment_type*
5	**Customer**	*dim_customer, dim_customer_segment, dim_customer_country, dim_customer_state, dim_customer_city*
3	**Product**	*dim_product, dim_category, dim_department*
5	**Order Destination**	*dim_order_country, dim_order_state, dim_order_region, dim_order_city, dim_market*
2	**Time**	*dim_date, dim_hour*

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Problem Statement



DataCo Global, an online retailer that sells clothing, sports gear, and electronics around the world. Their sales, shipping, and website data are stored separately, which makes it hard for managers to spot late deliveries, underperforming markets, or the impact of discounts and cancellations on profits. This project will create a data warehouse and an interactive dashboard to bring all this information together and track key performance indicators. With these tools, DataCo's managers will be able to find issues with delivery, profitability, and order quality sooner.

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Key Business Questions

01 How are the monthly sales trending, and which months show growth or decline?

02 Which markets, departments, and customer segments generate great sales and profit?

03 Which departments or products fall below the marginal target?

04 Does adding discounts reduce profit, and by how much?

05 What share of orders is late, how does it vary by shipping mode and market?

06 How different are the actual shipping days from the scheduled days?

07 What orders share the same characteristics of canceled or suspected fraud?

08 Which payment types and markets show higher cancellation or fraud rates?

09 How many customers repeat orders?

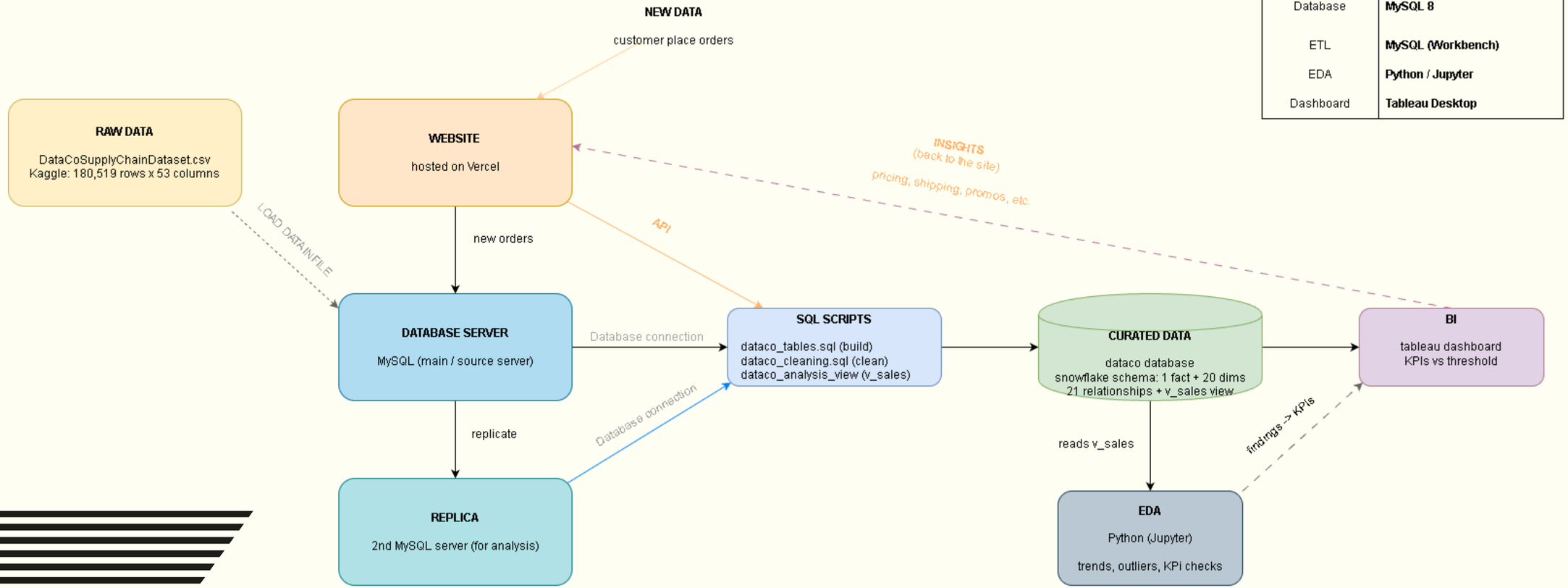
10 Do late deliveries reduce repeat orders?

Key Business Questions

KPI	Formula	Target	Warning	Purpose
Profit Margin	Total Order Profit / Total Sales	**12% or higher**	**Below 5%**	Shows whether the business sales are making positive money
Monthly Sales Growth	(Sales of current month – Sales of previous month) / Sales of previous month	**3% or higher**	**Below 0%**	Shows the business direction based on sales
Average Discount Rate	Total Discount / Gross Item Price	**10% or lower**	**Above 20%**	Shows how discounts affect the profit
Late Delivery Rate	Late Orders / All Orders	**15% or lower**	**Above 30%**	Shows how often late order occur from every order
Average Delivery Delay	Average of (Actual – Scheduled Shipping Days) per order	**0.5 days or lower**	**Above 1.5 days**	Shows how the actual delivery occurs from the scheduled delivery
Cancelled/Suspected Fraud Rate	Orders with Cancelled/Suspected Fraud Rate status / All Orders	**5% or lower**	**Above 8%**	Shows how much of the order quality are deemed cancelled/fraud
Repeat Customer Rate	Customer (2 or more orders) / all customers	**30% or higher**	**Below 15%**	Shows customer loyalty

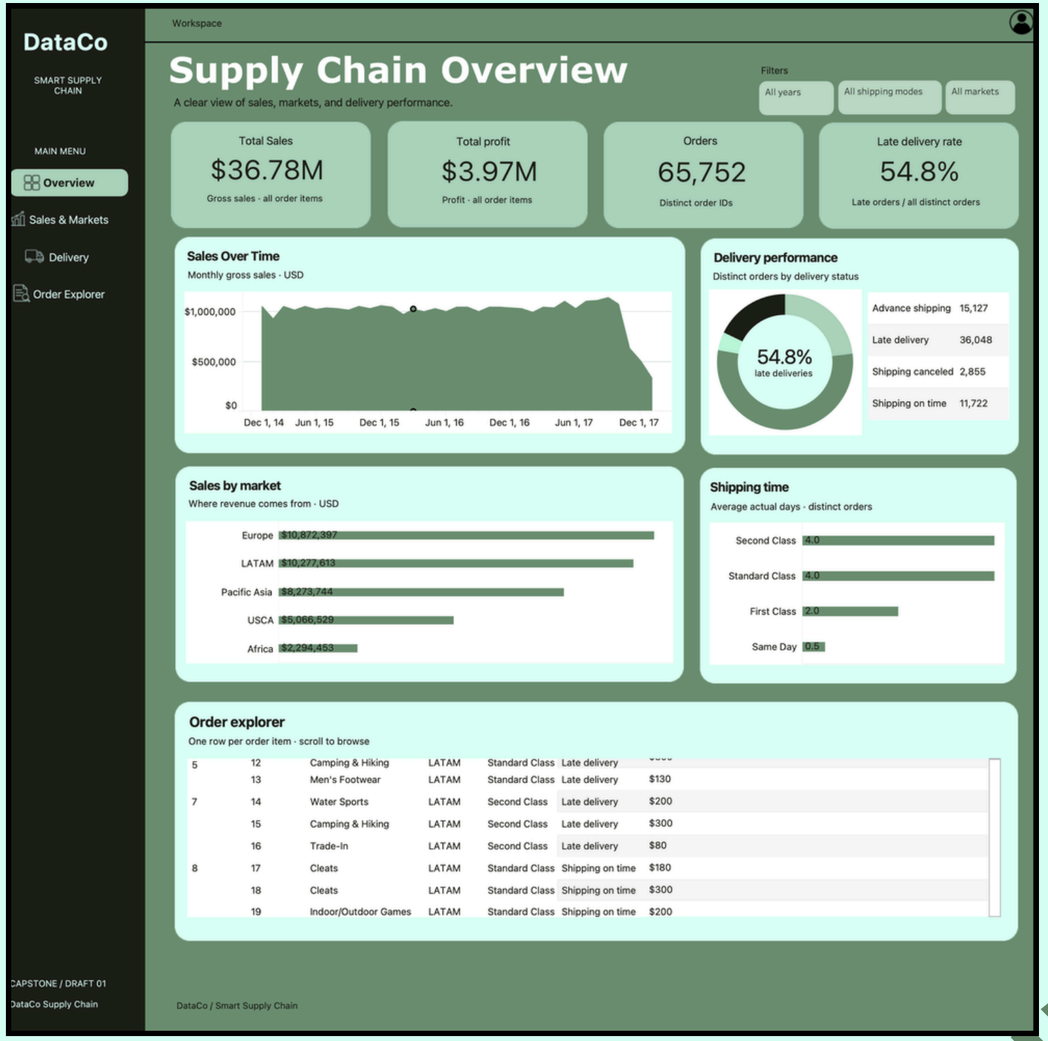
Solution Architecture

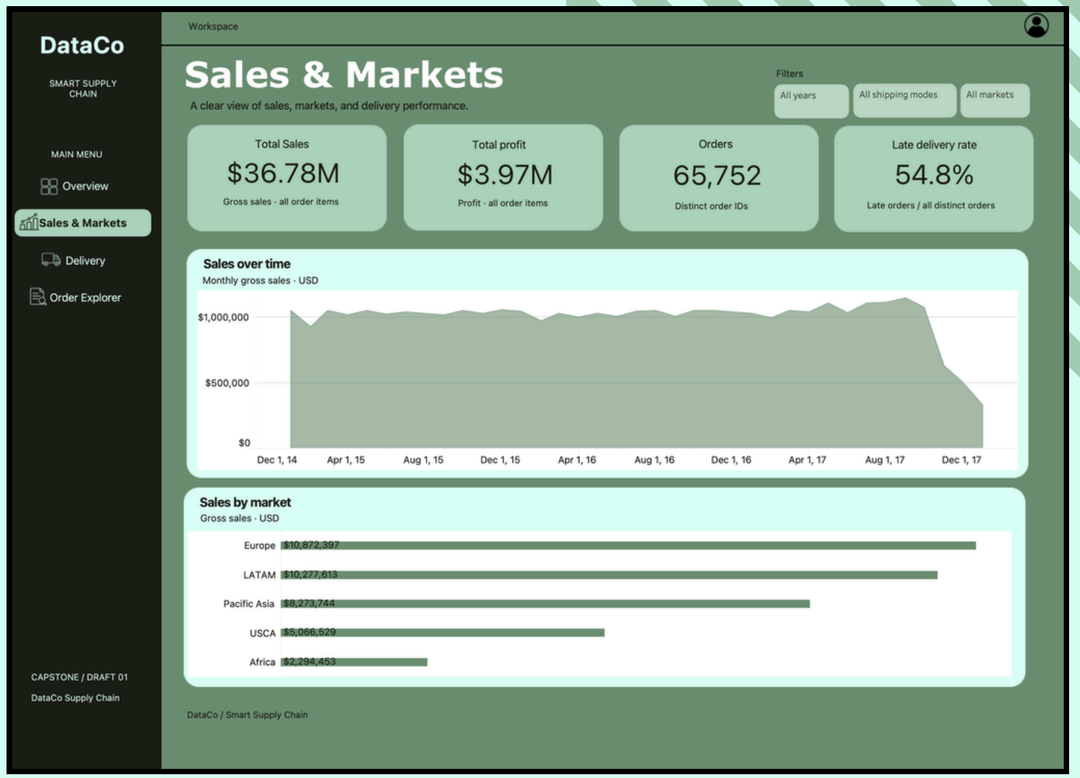
TECHSTACK	
Website	**Vercel (Canva mockup for now)_**
Database	**MySQL 8**
ETL	**MySQL (Workbench)**
EDA	**Python / Jupyter**
Dashboard	**Tableau Desktop**



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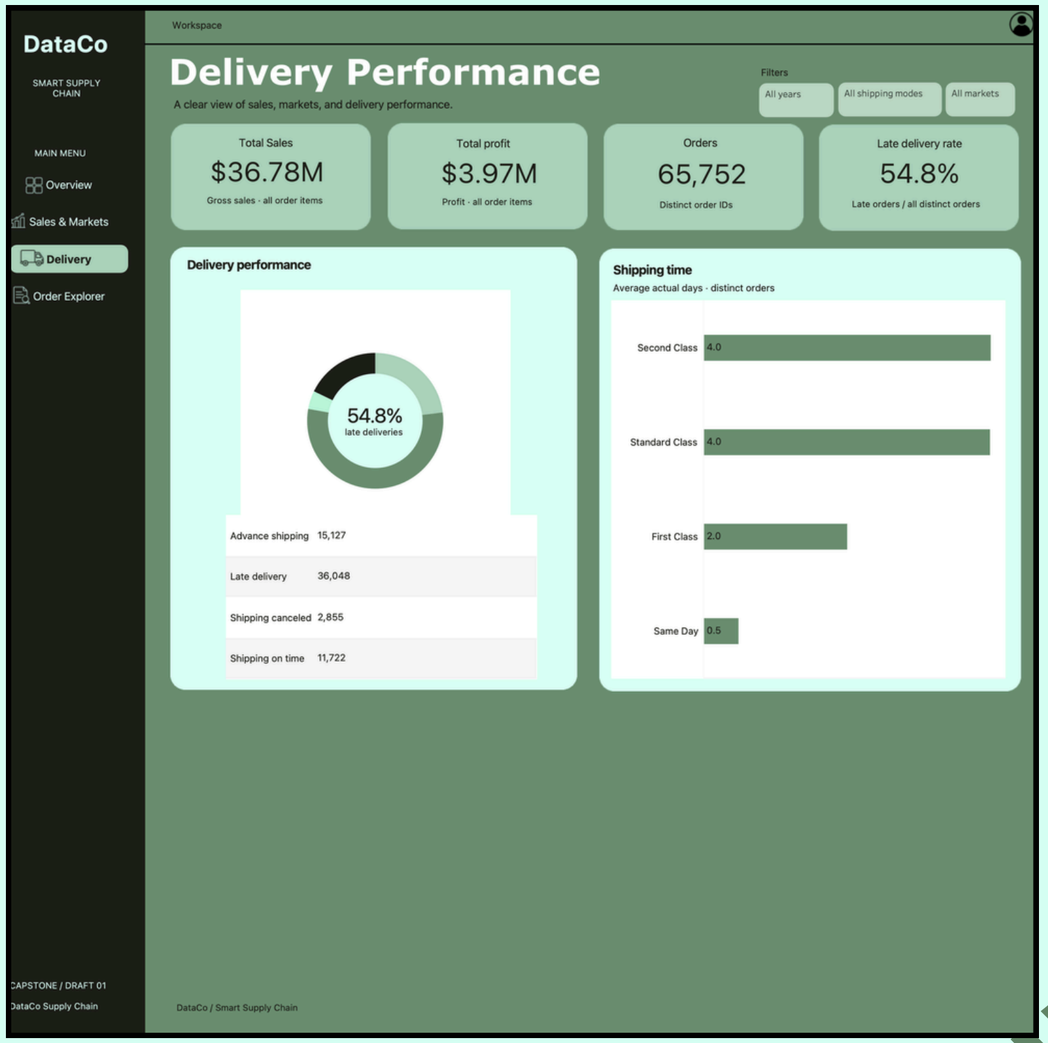
Dashboard Overview

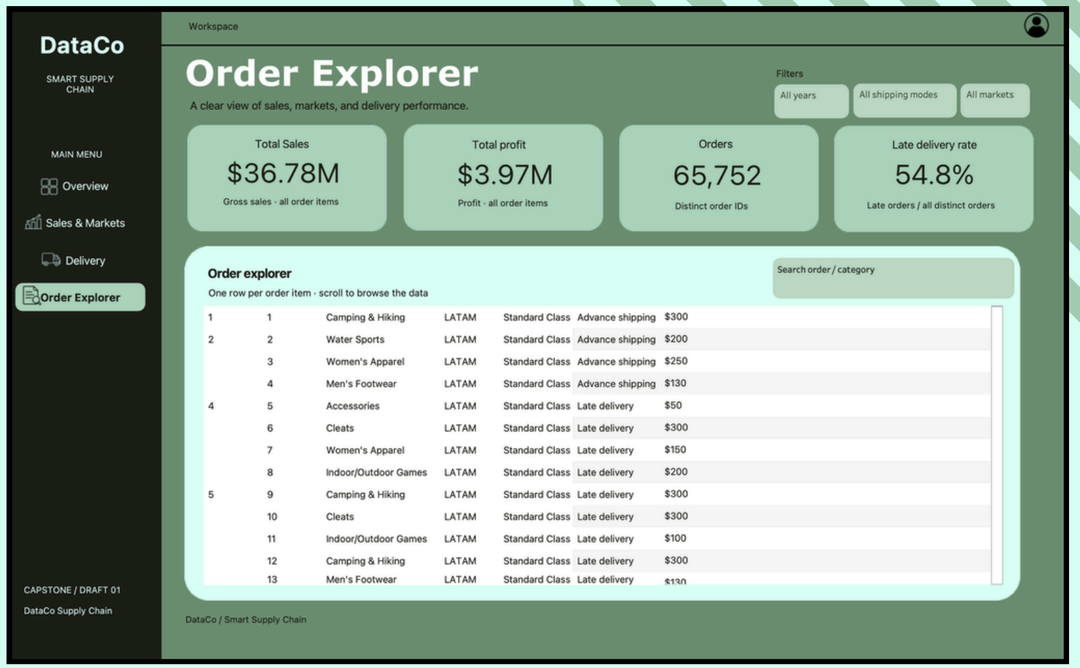




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Dashboard Overview





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Website UI/UX



WEBSITE LINK

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THANK YOU!

References:

Constante, F. (2019). DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS. Data Archiving and Networked Services (DANS), 5. https://doi.org/10.17632/8gx2fvg2k6

Tiwari, S. (2019, December 5). DataCo smart supply chain for big data analysis. Kaggledatasets. https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis